

Gaming Castle

Online Gaming Centre Management System

By

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Version History

Version No.	Date	Author(s)	Summary of Changes
01	2026/01/15	Kirisigan, Jashvanth, Thivya, Nirojan	Initial draft of SRS document
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Chapter 1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document is produced to define the functional and non-functional requirements for the Gaming Castle Online Gaming Centre Management System. A clear and comprehensive description of the system to be developed is provided by this document, and it is intended to serve as a contractual agreement between the development team and the client.

The intended audience of this document is identified as the project development team, academic evaluators, and the client (Gaming Castle management). The design, development, testing, and deployment phases of the system are guided by this document.

1.2 Scope

The system to be developed is named Gaming Castle, and it is designed as an Online Gaming Centre Management Web Application to support the operations of the Gaming Castle gaming hub located in Nellyyadi, Sri Lanka.

An integrated platform for managing slot bookings, tournament registrations, payment processing, customer loyalty programs, and administrative operations shall be provided by the system. PlayStation 5 (PS5) gaming slots shall be allowed to be booked online by registered customers, and participation in tournaments, as well as access to a loyalty reward system, shall be enabled.

Two primary user roles, namely Customer and Administrator, shall be supported by the system. Registration, login, gaming slot reservations, tournament participation, and the viewing of booking and reward history shall be enabled for customers. Booking management, tournament organisation and supervision, activity monitoring, and report generation for decision-making shall be carried out by administrators.

The system shall be developed as a web-based application, and accessibility through modern web browsers shall be ensured for both customers and administrators.

The development of games, game streaming services, integration with external gaming platforms such as the PlayStation Network, and hardware-level control of gaming consoles shall not be included in the system.

1.3 Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
Admin	System Administrator – Gaming Castle staff managing the platform
User / Customer	Registered individual who books slots or participates in tournaments
Slot	A reserved time block for using a PS5 console at Gaming Castle
Tournament	A competitive gaming event organised and managed through the system
Loyalty Program	A rewards scheme offering discounts to repeat customers
PS5	PlayStation 5
UI	User Interface
API	Application Programming Interface
IEEE	Institute of Electrical and Electronics Engineers
ISO/IEC 25010	International standard for software product quality characteristics
MVP	Minimum Viable Product
UAT	User Acceptance Testing

Table 2: Definitions, Acronyms and Abbreviations

1.4 Document Overview

This document is organised as follows:

- Chapter 1 – Introduction: Provides the purpose, scope, definitions, and overview of the document.
- Chapter 2 – Project Scope and Stakeholders: Describes project boundaries and identifies all key stakeholders.
- Chapter 3 – Functional Requirements: Lists all functional requirements grouped by system modules in tabular format.
- Chapter 4 – Non-Functional Requirements: Details measurable quality requirements categorised per ISO/IEC 25010.
- Chapter 5 – Business Rules, Constraints, and Assumptions: Documents operational rules and project limitations.
- Chapter 6 – Use Case Diagrams and Descriptions: Presents use cases for all system actors.
- Chapter 7 – Activity Diagrams: Illustrates key workflows within the system.
- Chapter 8 – External Interface Requirements: Specifies UI, hardware, software, and communication interfaces.
- Chapter 9 – Data Requirements: Defines the data entities and storage requirements.
- Chapter 10 – Alternative Solutions Considered: Discusses alternative approaches evaluated.
- Chapter 11 – Feasibility Study: Assesses technical, economic, operational, and schedule feasibility.
- Chapter 12 – Risk Analysis: Identifies and evaluates project risks with mitigation strategies.
- Chapter 13 – References: Lists all cited sources and standards.

Chapter 2 Project Scope and Stakeholders

2.1 Project Scope

Gaming Castle is a modern gaming hub located in Nellyyadi, Sri Lanka, where two PS5 consoles are operated and a selection of popular games is offered on a pay-as-you-go basis. Manual processes are currently relied upon for booking, payments, tournament management, and customer offers, resulting in operational inefficiencies.

All key operations of Gaming Castle shall be digitised and streamlined by the proposed web-based system. Online slot booking, automated payment management, structured tournament registration, a loyalty discount programme, and a comprehensive administrative dashboard shall be enabled by the system.

2.2 Stakeholders

Stakeholder	Role	Interest in System
Gaming Castle Owner	Client / Primary Stakeholder	Oversees overall business operations, monitors revenue, and ensures business growth and efficiency through the system.
Administrator (Staff)	System User – Admin Role	Manages bookings, tournaments, payments, and reports.
Registered Customers	System User – Customer Role	Books slots, joins tournaments, earns loyalty rewards.
Walk-in Customers	Indirect User	Uses the system indirectly through administrator-assisted booking of gaming slots.
Development Team	System Developers	Responsible for designing, developing, testing, and deploying the system according to requirements.
Project Supervisor	Academic Stakeholder	Provides academic oversight and evaluation.
University of Kelaniya	Academic Institution	Sets degree requirements and quality standards.

Table 3: Stakeholders

Chapter 3 Functional Requirements

3.1 User Registration and Authentication

ID	Description	Priority	Stakeholder	Status
FR-01	The system shall allow customers to register using their name, email address, password, and phone number.	High	Customer	Approved
FR-02	The system shall allow registered users to log in using their email and password.	High	Customer	Approved
FR-03	The system shall allow registered users to log in using their phone number and password.	High	Customer	Approved
FR-04	The system shall allow users to reset their passwords via email verification.	High	Customer	Approved
FR-05	The system shall allow users to reset their passwords via phone number verification.	High	Customer	Approved
FR-06	The system shall allow the administrator to manage user accounts (create, update, deactivate).	High	Admin	Approved

Table 4: Functional Requirements – User Registration and Authentication

3.2 Slot Booking Management

ID	Description	Priority	Stakeholder	Status
FR-07	The system shall display available time slots for PS5 consoles in real-time.	High	Customer	Approved
FR-08	The system shall allow registered customers to book a gaming slot by selecting a date, time, and console.	High	Customer	Approved
FR-09	The system shall allow the administrator to book slots on behalf of walk-in customers.	High	Admin	Approved

FR-10	The system shall prevent double-booking of the same console slot.	High	Customer	Approved
FR-11	The system shall allow customers to cancel or reschedule a booked slot within a defined time window.	High	Customer	Approved
FR-12	The system shall send booking confirmation notifications to customers via email or SMS.	Medium	Customer	Approved

Table 5: Functional Requirements – Slot Booking Management

3.3 Tournament Management

ID	Description	Priority	Stakeholder	Status
FR-13	The system shall allow the administrator to create and publish tournament events with details (game, date, entry fee, max participants).	High	Admin	Approved
FR-14	The system shall allow registered customers to view and register for upcoming tournaments.	High	Customer	Approved
FR-15	The system shall manage tournament brackets and participant lists.	Medium	Admin	Approved
FR-16	The system shall notify registered participants of tournament schedules and results.	Medium	User	Approved
FR-17	The system shall allow the administrator to update tournament status (upcoming, ongoing, completed).	Medium	Admin	Approved
FR-18	The system shall include an automated tournament management module that generates tournament brackets and manages match progression based on predefined logic or historical data.	Medium	Admin	Approved

Table 6: Functional Requirements – Tournament Management

3.4 Payment Management

ID	Description	Priority	Stakeholder	Status
FR-19	The system shall calculate charges based on booked hours at the applicable hourly rate.	High	Customer	Approved
FR-20	The system shall support online payment processing for slot bookings and tournament entry fees.	High	Customer	Approved
FR-21	The system shall generate and display digital receipts upon successful payment.	High	Customer	Approved
FR-22	The system shall allow the administrator to record cash payments for walk-in customers.	High	Admin	Approved
FR-23	The system shall provide a payment history log accessible to customers and the administrator.	Medium	Admin / Customer	Approved

Table 7: Functional Requirements – Payment Management

3.5 Loyalty Program

ID	Description	Priority	Stakeholder	Status
FR-24	The system shall award loyalty points to customers for each completed booking.	High	Customer	Approved
FR-25	The system shall allow customers to redeem loyalty points for discounts on future bookings.	High	Customer	Approved
FR-26	The system shall display a customer's current loyalty points balance on their profile.	Medium	Customer	Approved
FR-27	The system shall allow the administrator to configure loyalty point earning and redemption rules.	Medium	Admin	Approved

Table 8: Functional Requirements – Loyalty Program

3.6 Administrative Dashboard

ID	Description	Priority	Stakeholder	Status
FR-28	The system shall provide an administrative dashboard displaying real-time booking status, revenue summary, and active tournaments.	High	Admin	Approved
FR-29	The system shall allow the administrator to generate reports on bookings, revenue, and tournament participation.	High	Admin	Approved
FR-30	The system shall allow the administrator to manage game titles available for booking.	Medium	Admin	Approved
FR-31	The system shall allow the administrator to view and manage all registered customer accounts.	Medium	Admin	Approved

Table 9: Functional Requirements – Administrative Dashboard

Chapter 4 Non-Functional Requirements

All non-functional requirements are categorised in accordance with ISO/IEC 25010 and include measurable acceptance criteria.

4.1 Performance

ID	Description	Acceptance Criterion
NFR-01	The system shall respond to all page load requests efficiently.	95% of pages shall load within 3 seconds under normal load conditions.

NFR-02	The system shall handle concurrent users without degradation.	The system shall support at least 100 concurrent users without a measurable performance drop.
NFR-03	Slot availability updates shall be reflected in near real-time.	Slot status updates shall be visible to all users within 5 seconds of a change.

Table 10: Non-Functional Requirements – Performance

4.2 Reliability

ID	Description	Acceptance Criterion
NFR-04	The system shall maintain high availability.	The system shall achieve a minimum uptime of 99% per month, excluding planned maintenance windows.
NFR-05	The system shall recover from failures gracefully.	In the event of an unexpected failure, the system shall restore full system functionality within 30 minutes.

Table 11: Non-Functional Requirements – Reliability

4.3 Usability

ID	Description	Acceptance Criterion
NFR-06	The system shall be easy to use for first-time users.	A first-time user shall be able to complete a slot booking within 5 minutes without external assistance.
NFR-07	The system shall support responsive design.	The system shall render correctly on desktop, tablet, and mobile screen sizes (minimum 320px width).
NFR-08	The system shall display error messages clearly.	All user-facing error messages shall be written in plain English and suggest a corrective action.

Table 12: Non-Functional Requirements – Usability

4.4 Security

ID	Description	Acceptance Criterion
NFR-09	All data transmissions shall be encrypted.	The system shall enforce HTTPS (TLS 1.2 or higher) for all communications.
NFR-10	User passwords shall be stored securely.	Passwords shall be hashed using bcrypt or an equivalent algorithm; plain-text storage is prohibited.
NFR-11	The system shall enforce role-based access control.	Administrative functions shall be accessible only to authenticated admin accounts; customer accounts shall not access admin panels.
NFR-12	The system shall protect against common web vulnerabilities.	The system shall pass an OWASP Top 10 vulnerability assessment prior to production deployment.

Table 13: Non-Functional Requirements – Security

4.5 Scalability

ID	Description	Acceptance Criterion
NFR-13	The system shall be designed to accommodate future growth.	The system architecture shall support horizontal scaling to handle a 3x increase in user load without a redesign.
NFR-14	The database shall handle increasing data volumes efficiently.	Query response times shall remain within 2 seconds under normal conditions as the database grows to 100,000 booking records.

Table 14: Non-Functional Requirements – Scalability

4.6 Maintainability

ID	Description	Acceptance Criterion
NFR-15	The codebase shall be well-documented.	All modules, functions, and APIs shall include inline comments and a README; documentation coverage shall exceed 80%.

NFR-16	The system shall support modular architecture.	Each major feature (booking, payments, tournaments, loyalty) shall be implemented as an independently maintainable module.
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Table 15: Non-Functional Requirements – Maintainability

4.7 Portability

ID	Description	Acceptance Criterion
NFR-17	The system shall operate across major web browsers.	The system shall function correctly on the latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
NFR-18	The system shall be deployable on standard cloud infrastructure.	The system shall be deployable on platforms such as AWS, Google Cloud, or Azure with minimal configuration changes.

Table 16: Non-Functional Requirements – Portability

Chapter 5 Business Rules, Constraints, and Assumptions

5.1 Business Rules

- BR-01: A customer must be registered and logged in to make an online slot booking.
- BR-02: Each PS5 console may only be assigned to one customer per time slot.
- BR-03: Slot bookings shall be available in minimum increments of one hour.
- BR-04: Cancellations made more than two hours before the slot start time shall be eligible for a full refund or rescheduling.
- BR-05: Loyalty points shall be awarded only upon completion and confirmation of a paid booking.
- BR-06: Tournament registration shall close once the maximum participant limit is reached or the registration deadline passes.
- BR-07: Override capability shall be provided to the administrator to allow any booking to be cancelled or reassigned.
- BR-08: Walk-in customer bookings recorded by the admin shall default to cash payment.

5.2 Constraints

- The system must be developed as a web application accessible via standard web browsers.
- The initial deployment shall support 2 PS5 consoles; the system shall be scalable to accommodate additional consoles in the future.
- The project must be completed within the academic semester timeline.
- The development team consists of 4 members; resource allocation is constrained accordingly.
- The system shall be built using open-source or freely available frameworks to minimise licensing costs.
- Internet connectivity is required for the system to function; offline mode is not within the project scope.

5.3 Assumptions

- It is assumed that Gaming Castle management will provide timely feedback and sign-off at each project milestone.
- It is assumed that all customers have access to a device with internet connectivity and a modern web browser.
- It is assumed that the hourly rates and game titles will be provided by the client prior to system development.
- It is assumed that a payment gateway (e.g., PayHere or Stripe) will be available and accessible for integration.
- It is assumed that the Gaming Castle premises will have stable internet connectivity for admin operations.
- It is assumed that sufficient predefined rules or sample data will be available to configure or simulate the automated tournament management module.

Chapter 6 Use Case Diagrams and Descriptions

6.1 System Actors

The Gaming Castle system involves two primary actors:

- Customer (Registered User): A person who accesses the system online to book slots, register for tournaments, manage their profile, and redeem loyalty rewards.

- Administrator (Admin): A member of Gaming Castle staff who manages bookings, tournaments, payments, customer accounts, and shop operations through the admin dashboard.

6.2 Use Case Summary

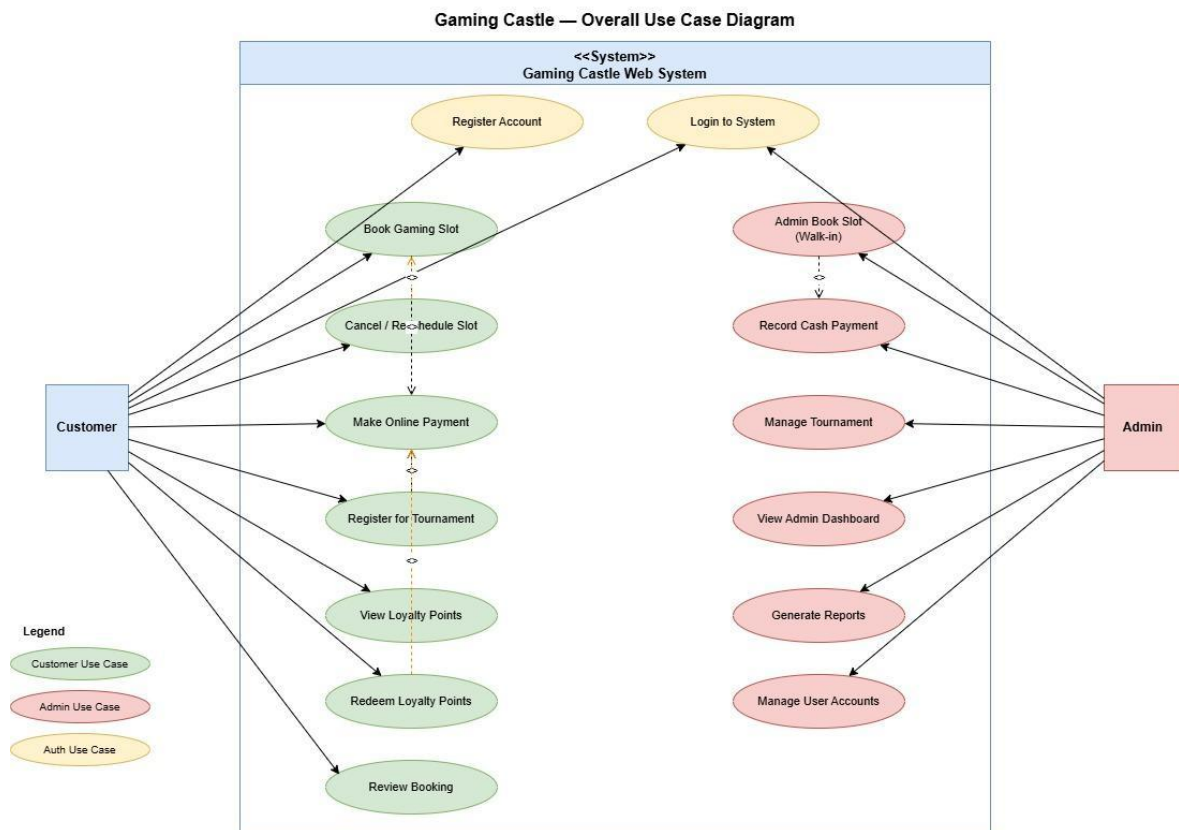


Figure 1: Overall Use Case Diagram – Gaming Castle Web System

Use Case ID	Use Case Name	Actor(s)	Priority
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UC-01	Register Account	Customer	High
UC-02	Login to the System	Customer / Admin	High
UC-03	Book Gaming Slot	Customer	High
UC-04	Cancel / Reschedule Slot	Customer	Medium
UC-05	Admin Book Slot for Walk-in Customer	Admin	High
UC-06	Make Online Payment	Customer	High
UC-07	Record Cash Payment	Admin	High
UC-08	Register for Tournament	Customer	High
UC-09	Manage Tournament	Admin	High
UC-10	View Loyalty Points	Customer	Medium
UC-11	Redeem Loyalty Points	Customer	Medium
UC-12	View Admin Dashboard	Admin	High
UC-13	Generate Reports	Admin	Medium
UC-14	Manage User Accounts	Admin	Medium

Table 17: Use Case Summary

6.3 Use Case Descriptions

Use Case UC-01: Register Account

Attribute	Details
Use Case ID	UC-01
Use Case Name	Register Account
Actor(s)	Customer
Preconditions	The customer has not previously registered. The system is accessible.
Main Flow	1. The customer navigates to the Register page. 2. Required details are entered (name, email, phone number, password). 3. The input fields are validated by the system. 4. It is verified by the system that the email or phone number is not already registered.

	5. The account is created and the details are stored by the system. 6. A success message is displayed and the user is redirected to the Login page.
Alternative Flow	3a. If any field is invalid – the error is highlighted and correction is prompted by the system. 4a. If the email or phone number is already registered – the following message is displayed: An account with this email already exists.
Postconditions	A new customer account is created and stored in the system. The customer may now log in.
Exception Flow	If the system is unavailable, a service error message is displayed and the customer is prompted to try again later.
Related Requirements	FR-01, FR-02, FR-03

Table 18: Use Case Description – UC-01: Register Account

Use Case UC-02: Login to System

Attribute	Details
Use Case ID	UC-02
Use Case Name	Login to System
Actor(s)	Customer / Admin
Preconditions	The user has a registered account. The system is accessible.
Main Flow	1. The user navigates to the Login page. 2. An email address and password are entered by the user. 3. The credentials are validated by the system. 4. The user is authenticated and the role (Customer or Admin) is identified by the system. 5. The user is redirected to the appropriate dashboard (Customer home or Admin dashboard).
Alternative Flow	3a. If credentials are invalid – an error message is displayed and the user is prompted to re-enter. 3b. If the password has been forgotten – the user selects Forgot Password and is redirected to the password reset flow.
Postconditions	The user is authenticated and granted access to role-appropriate features.

Exception Flow	If the account is deactivated, the following message is displayed: Your account has been deactivated. Please contact support.
Related Requirements	FR-02, FR-03, FR-04, FR-05

Table 19: Use Case Description – UC-02: Login to System

Use Case UC-03: Book Gaming Slot

Attribute	Details
Use Case ID	UC-03
Use Case Name	Book Gaming Slot
Actor(s)	Customer
Preconditions	The customer is registered and logged into the system.
Main Flow	1. The customer navigates to the Book a Slot page. 2. The preferred date and time are selected. 3. An available PS5 console and game are selected. 4. The booking summary with the applicable charge is displayed by the system. 5. The booking is confirmed and payment is proceeded with by the customer. 6. Payment is processed and the booking is confirmed by the system. 7. A confirmation notification is sent to the customer.
Alternative Flow	If no slots are available for the selected time, the customer is notified and alternative times are suggested by the system.
Postconditions	A confirmed booking record is created. The slot is marked as unavailable for other users.
Exception Flow	If payment fails, the slot is released and the customer is notified.
Related Requirements	FR-07, FR-08, FR-10, FR-12, FR-19, FR-20, FR-21

Table 20: Use Case Description – UC-03: Book Gaming Slot

Use Case UC-04: Cancel / Reschedule Slot

Attribute	Details
Use Case ID	UC-04

Use Case Name	Cancel / Reschedule Slot
Actor(s)	Customer
Preconditions	The customer is registered and logged in. An existing confirmed booking is held by the customer.
Main Flow	1. The customer navigates to My Bookings. 2. A booking to cancel or reschedule is selected. 3. The system checks whether the cancellation/reschedule window is still open (more than 2 hours before slot start). 4. The cancellation is confirmed or a new date and time is selected by the customer. 5. The booking record is updated accordingly by the system. 6. The customer is notified of the change via email or SMS.
Alternative Flow	4a. If rescheduling – available slots are displayed for the customer to choose from. 4b. If cancelling – a refund is processed by the system if applicable.
Postconditions	The original slot is released and made available for other users. The booking record is updated or cancelled.
Exception Flow	If the cancellation window has passed (less than 2 hours before the slot), the customer is informed that the booking can no longer be modified.
Related Requirements	FR-11, FR-12

Table 21: Use Case Description – UC-04: Cancel / Reschedule Slot

Use Case UC-05: Admin Book Slot for Walk-in Customer

Attribute	Details
Use Case ID	UC-05
Use Case Name	Admin Book Slot for Walk-in Customer
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access. At least one slot is available.
Main Flow	1. The admin navigates to the Admin Booking panel. 2. An available date, time, console, and game are selected. 3. The walk-in customer details are entered by the admin. 4. The booking is created and the slot is marked as occupied by the system.

	5. The cash payment is recorded by the admin. 6. Revenue records are updated and a receipt is generated by the system.
Alternative Flow	2a. If no slots are available for the selected time – the admin is notified and alternative times are suggested.
Postconditions	A booking record is created for the walk-in customer. The slot is marked as occupied. Cash payment is recorded.
Exception Flow	If the system fails to save the booking, an error message is displayed and the admin is prompted to retry.
Related Requirements	FR-09, FR-22

Table 22: Use Case Description – UC-05: Admin Book Slot for Walk-in Customer

Use Case UC-06: Make Online Payment

Attribute	Details
Use Case ID	UC-06
Use Case Name	Make Online Payment
Actor(s)	Customer
Preconditions	The customer is registered and logged in. A pending booking awaiting payment exists.
Main Flow	1. The booking summary with the total charge is displayed by the system. 2. An online payment method (card or gateway) is selected by the customer. 3. Payment details are entered and confirmed by the customer. 4. The payment is processed via the integrated payment gateway. 5. The payment is confirmed and the booking is finalised. 6. A digital receipt is generated and displayed. 7. A booking confirmation notification is sent to the customer.
Alternative Flow	3a. If the customer cancels the payment – the system returns to the booking summary and the slot remains reserved briefly before being released.
Postconditions	Payment is confirmed. The booking is finalised. Loyalty points are awarded to the customer.

Exception Flow	If the payment fails (e.g. insufficient funds, gateway error) – the customer is notified, the slot reservation is released, and a retry is prompted.
Related Requirements	FR-19, FR-20, FR-22, FR-23,FR-24

Table 23: Use Case Description – UC-06: Make Online Payment

Use Case UC-07: Record Cash Payment

Attribute	Details
Use Case ID	UC-07
Use Case Name	Record Cash Payment
Actor(s)	Admin
Preconditions	The admin is authenticated. A walk-in booking has been created by the admin.
Main Flow	1. The admin navigates to the payment recording section. 2. The relevant booking is selected. 3. The cash amount received is entered by the admin. 4. The amount is validated by the system against the booking charge. 5. The booking is marked as paid by the system. 6. A receipt is generated and revenue records are updated.
Alternative Flow	3a. If the amount entered does not match the expected charge – the admin is alerted to confirm before proceeding.
Postconditions	The booking is marked as paid (cash). Revenue records are updated. A receipt is generated.
Exception Flow	If the system fails to record the payment, an error is displayed and the admin is prompted to retry.
Related Requirements	FR-21, FR-22

Table 24: Use Case Description – UC-07: Record Cash Payment

Use Case UC-08: Register for Tournament

Attribute	Details
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Use Case ID	UC-08
Use Case Name	Register for Tournament
Actor(s)	Customer
Preconditions	The customer is registered and logged in. A tournament with open registration exists.
Main Flow	1. The customer navigates to the Tournaments page. 2. Available upcoming tournaments are viewed. 3. A tournament is selected and its details are viewed (game, date, entry fee, max participants). 4. The customer selects Register and is directed to payment. 5. Payment of the entry fee is completed. 6. Registration is confirmed and the customer is added to the participant list. 7. A confirmation notification is sent to the customer.
Alternative Flow	5a. If payment fails – registration is not confirmed and the customer is notified.
Postconditions	The customer is registered for the tournament. Entry fee payment is recorded.
Exception Flow	If the tournament is full or the registration deadline has passed – the customer is informed that registration is closed.
Related Requirements	FR-14, FR-19

Table 25: Use Case Description – UC-08: Register for Tournament

Use Case UC-09: Manage Tournament

Attribute	Details
Use Case ID	UC-09
Use Case Name	Manage Tournament
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin navigates to the Tournament Management section. 2. A new tournament is created by entering details (name, game, date, entry fee, max participants).

	3. The tournament is published and made visible to customers by the system. 4. Registrations are monitored in real-time by the admin. 5. Tournament status is updated as it progresses. 6. Tournament results are recorded and published.
Alternative Flow	The admin may cancel a tournament before it begins; all registered participants are notified automatically by the system.
Postconditions	Tournament records are created/updated in the system.
Exception Flow	If the maximum participant limit is reached, registration is automatically closed by the system.
Related Requirements	FR-13, FR-15, FR-16, FR-17

Table 26: Use Case Description – UC-09: Manage Tournament

Use Case UC-10: View Loyalty Points

Attribute	Details
Use Case ID	UC-10
Use Case Name	View Loyalty Points
Actor(s)	Customer
Preconditions	The customer is registered and logged in.
Main Flow	1. The customer navigates to their Profile or Loyalty Rewards section. 2. The loyalty points balance is retrieved by the system. 3. The current points balance and a history of earned and redeemed points are displayed.
Alternative Flow	3a. If the customer has no points history – a balance of 0 is displayed along with a message encouraging the first booking.
Postconditions	The current loyalty points balance and transaction history are viewed by the customer.
Exception Flow	If the loyalty data cannot be loaded, an error is displayed and the customer is prompted to refresh the page.
Related Requirements	FR-25

Table 27: Use Case Description – UC-10: View Loyalty Points

Use Case UC-11: Redeem Loyalty Points

Attribute	Details
Use Case ID	UC-11
Use Case Name	Redeem Loyalty Points
Actor(s)	Customer
Preconditions	The customer is registered and logged in. Sufficient loyalty points for redemption are available.
Main Flow	1. The customer navigates to the Slot Booking page and proceeds to payment. 2. The available loyalty points and equivalent discount are displayed by the system. 3. The customer opts to redeem points against the booking charge. 4. The discount is applied and the updated total is displayed . 5. Payment for the remaining balance is confirmed and completed. 6. The redeemed points are deducted from the customer's balance.
Alternative Flow	3a. If redemption is not opted for – the booking proceeds at full price with no points deducted.
Postconditions	Loyalty points are deducted. The booking charge is reduced by the redeemed amount.
Exception Flow	If the points balance drops below the minimum redemption threshold during checkout, the option is disabled and the customer is notified.
Related Requirements	FR-24, FR-25

Table 28: Use Case Description – UC-11: Redeem Loyalty Points

Use Case UC-12: View Admin Dashboard

Attribute	Details
Use Case ID	UC-12

Use Case Name	View Admin Dashboard
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin logs in and is redirected to the Admin Dashboard. 2. Real-time summary widgets are displayed: current active tournaments, today's bookings, revenue summary, and upcoming tournaments. 3. The admin navigates to detailed sections (Bookings, Payments, Tournaments, Users, Reports) from the dashboard.
Alternative Flow	2a. If there are no active sessions or bookings – empty state messages are displayed for the relevant widgets.
Postconditions	A real-time operational overview of the Gaming Castle system is viewed by the admin.
Exception Flow	If real-time data fails to load, the last cached data is displayed with a warning that data may not be current.
Related Requirements	FR-27

Table 29: Use Case Description – UC-12: View Admin Dashboard

Use Case UC-13: Generate Reports

Attribute	Details
Use Case ID	UC-13
Use Case Name	Generate Reports
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin navigates to the Reports section from the Admin Dashboard. 2. A report type is selected (Bookings, Revenue, Tournament Participation). 3. The date range and any filters are specified. 4. Relevant data is retrieved and processed by the system. 5. The report is displayed in a structured format. 6. The report may be exported (e.g., PDF or CSV) by the admin.

Alternative Flow	5a. If no data is available for the selected criteria – the admin is informed that no records match the filter.
Postconditions	A report is generated and optionally exported by the admin.
Exception Flow	If report generation fails due to a system error, the admin is notified and prompted to retry.
Related Requirements	FR-28

Table 30: Use Case Description – UC-13: Generate Reports

Use Case UC-14: Manage User Accounts

Attribute	Details
Use Case ID	UC-14
Use Case Name	Manage User Accounts
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin navigates to the User Management section. 2. A list of all registered customer accounts is displayed. 3. Users may be searched or filtered by name, email, or status. 4. A user is selected to view their profile, booking history, and loyalty points. 5. Account details may be updated or the account may be deactivated if required. 6. Changes are saved and reflected immediately by the system.
Alternative Flow	5a. If a user is deactivated – a notification is sent to the affected user by the system.
Postconditions	User account details are updated or the account is deactivated as required.
Exception Flow	If the system fails to save changes, an error is displayed and the admin is prompted to retry.
Related Requirements	FR-06, FR-30

Table 31: Use Case Description – UC-14: Manage User Accounts

Chapter 7 Activity Diagrams

This chapter presents eight key workflows within the Gaming Castle system. Each activity diagram illustrates the sequence of activities, decision points, and possible outcomes for a primary use case.

7.1 Slot Booking

This activity diagram describes the process by which a registered customer books a PS5 gaming slot through the Gaming Castle web system. The workflow covers slot availability checking, booking confirmation, and payment processing.

Step	Activity	Actor	Decision / Outcome
1	Customer Logs In	Customer	Precondition: Customer must be registered
2	Navigate to Book a Slot Page	Customer	–
3	Select Date & Time	Customer	–
4	Check Slot Availability	System	Decision: Slot Available? YES → Step 5 NO → Show message, suggest alternatives → END
5	Select Console & Game	Customer	–
6	View Booking Summary & Charge	System	–
7	Confirm Booking	Customer	–
8	Proceed to Payment	System	–
9	Process Payment	System	Decision: Payment Successful? YES → Step 10 NO → Release slot, notify customer → END
10	Booking Confirmed	System	–

11	Send Confirmation Notification	System	Postcondition: Slot marked unavailable
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Table 32: Activity Steps – Slot Booking

Activity Diagram — Slot Booking

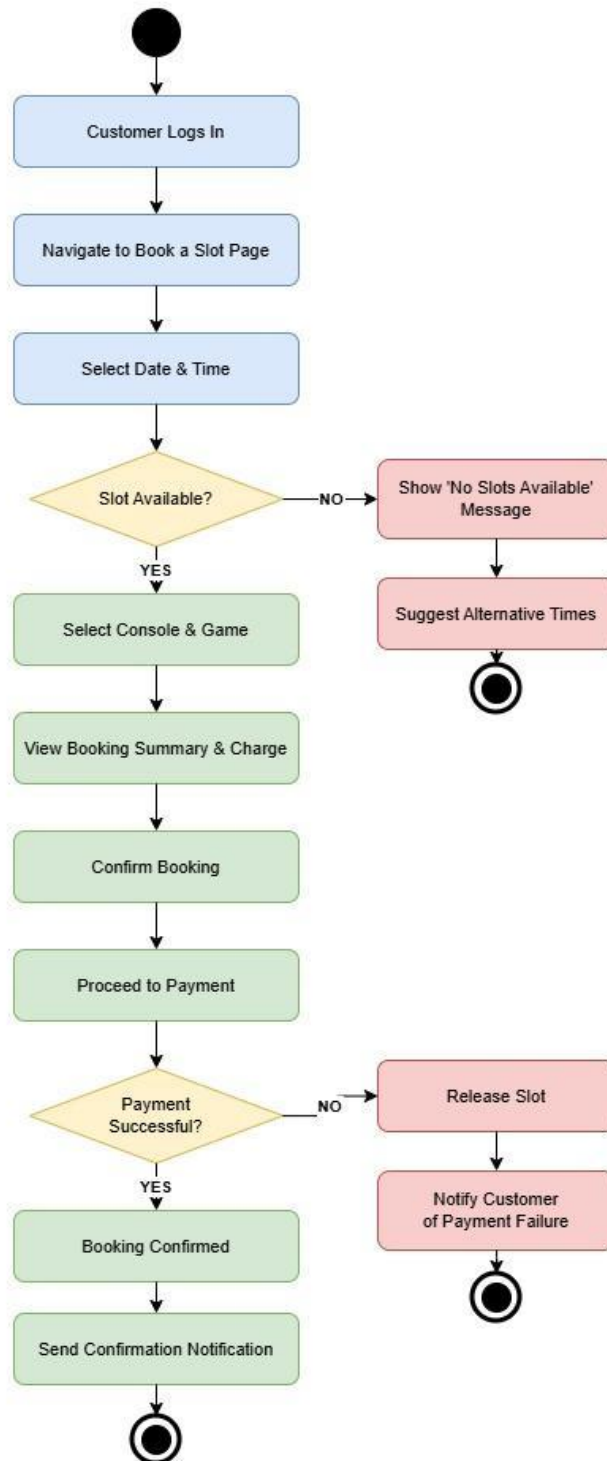


Figure 2: Activity Diagram – Slot Booking

7.2 Tournament Registration

This activity diagram describes the process by which a registered customer views, selects, and registers for a tournament on the Gaming Castle platform, including entry fee payment validation.

Step	Activity	Actor	Decision / Outcome
1	Customer Logs In	Customer	Precondition: Customer must be registered
2	Navigate to Tournaments Page	Customer	–
3	View Available Tournaments	Customer	–
4	Select a Tournament	Customer	–
5	Check Registration Status	System	Decision: Registration Open? NO → Show Closed message → END
6	Check Participant Limit	System	Decision: Tournament Full? YES → Show Full message → END
7	Click Register	Customer	–
8	Proceed to Entry Fee Payment	System	–
9	Process Payment	System	Decision: Payment Successful? YES → Step 10 NO → Notify customer → END
10	Registration Confirmed	System	–
11	Send Confirmation Notification	System	Postcondition: Participant count incremented

Table 33: Activity Steps – Tournament Registration

Activity Diagram — Tournament Registration

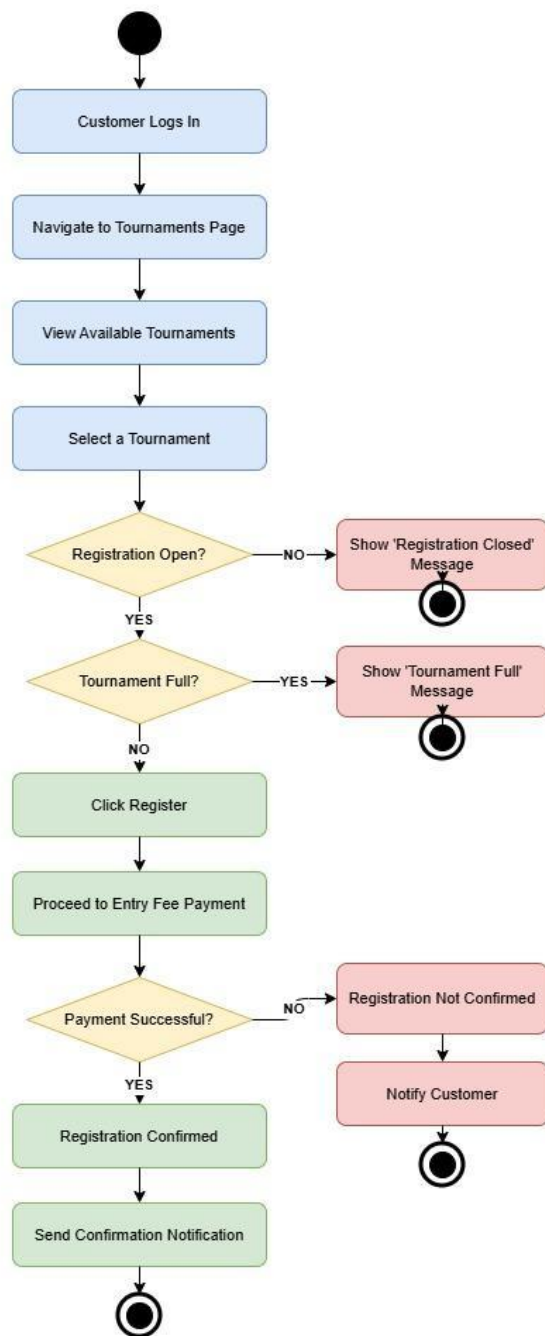


Figure 3: Activity Diagram – Tournament Registration

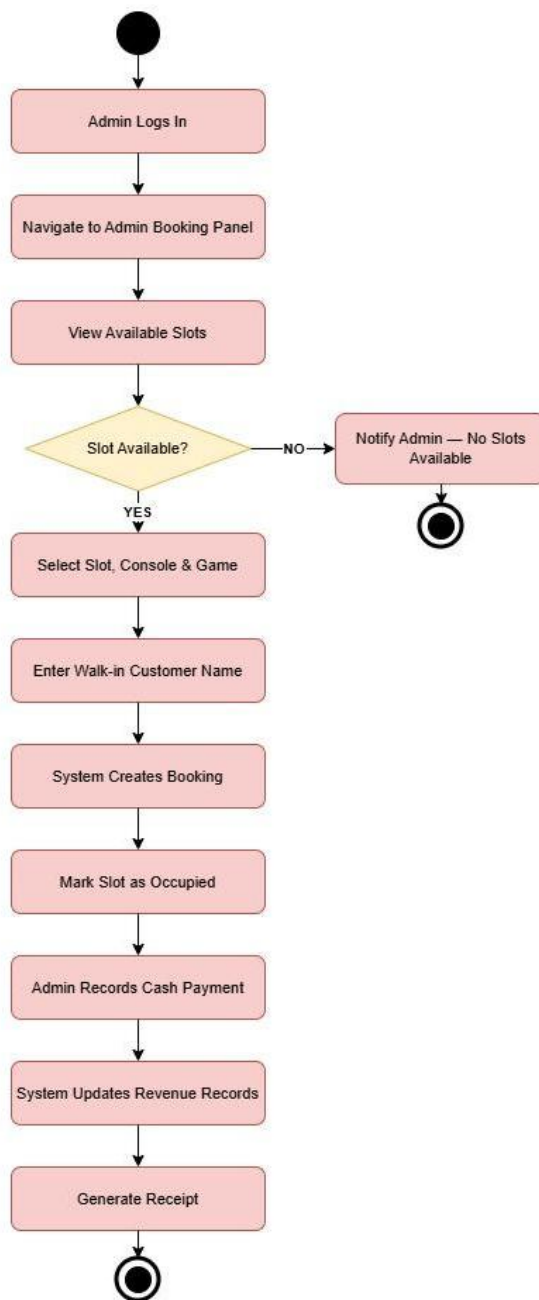
7.3 Admin Walk-in Booking

This activity diagram describes the process by which an administrator books a PS5 gaming slot on behalf of a walk-in customer who does not use the online system, and records a cash payment.

Step	Activity	Actor	Decision / Outcome
1	Admin Logs In	Admin	Precondition: Admin must be authenticated
2	Navigate to Admin Booking Panel	Admin	–
3	View Available Slots	Admin	–
4	Check Slot Availability	System	Decision: Slot Available? NO → Notify admin → END
5	Select Slot, Console & Game	Admin	–
6	Enter Walk-in Customer Name	Admin	–
7	System Creates Booking	System	–
8	Mark Slot as Occupied	System	–
9	Admin Records Cash Payment	Admin	–
10	System Updates Revenue Records	System	–
11	Generate Receipt	System	Postcondition: Revenue records updated

Table 34: Activity Steps – Admin Walk-in Booking

Activity Diagram — Admin Walk-in Booking



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Figure 4: Activity Diagram – Admin Walk-in Booking

7.4 Loyalty Points Redemption

This activity diagram describes the process by which a registered customer applies accumulated loyalty points as a discount during a slot booking payment.

Step	Activity	Actor	Decision / Outcome
1	Customer Logs In	Customer	–
2	Navigate to Book a Slot	Customer	–
3	View Loyalty Points Balance	System	–
4	Check Sufficient Points	System	Decision: Sufficient Points? NO → Proceed to full payment → END
5	Apply Loyalty Points Discount	System	–
6	View Discounted Booking Summary	Customer	–
7	Confirm & Pay Remaining Balance	Customer	–
8	Process Payment	System	Decision: Payment Successful? YES → Step 9 NO → Restore points, notify → END
9	Deduct Redeemed Points from Balance	System	–
10	Booking Confirmed	System	Postcondition: Points balance updated

Table 35: Activity Steps – Loyalty Points Redemption

Activity Diagram — Loyalty Points Redemption

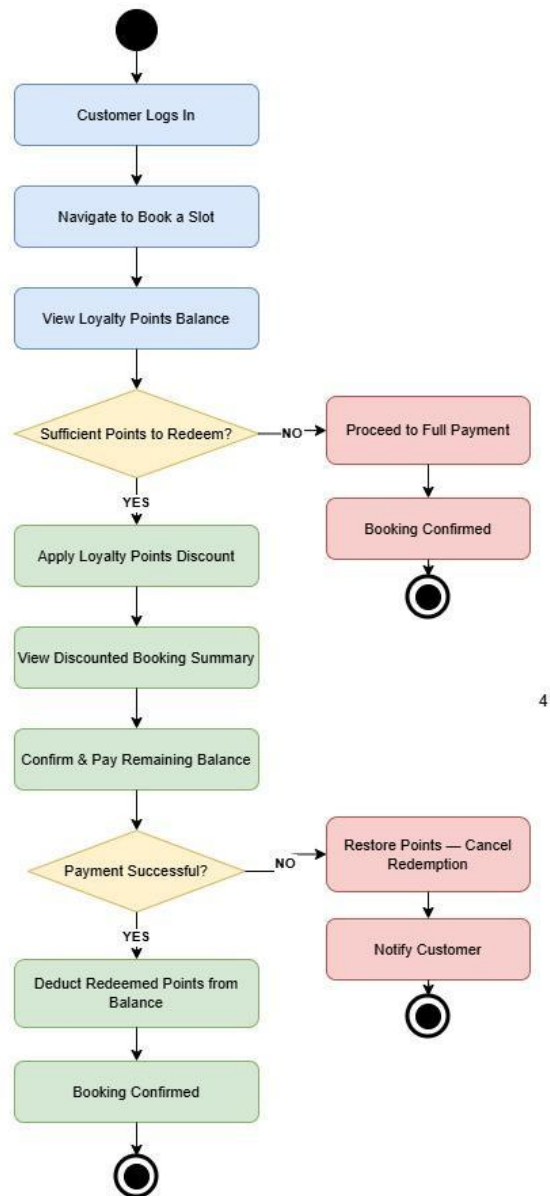


Figure 5: Activity Diagram – Loyalty Points Redemption

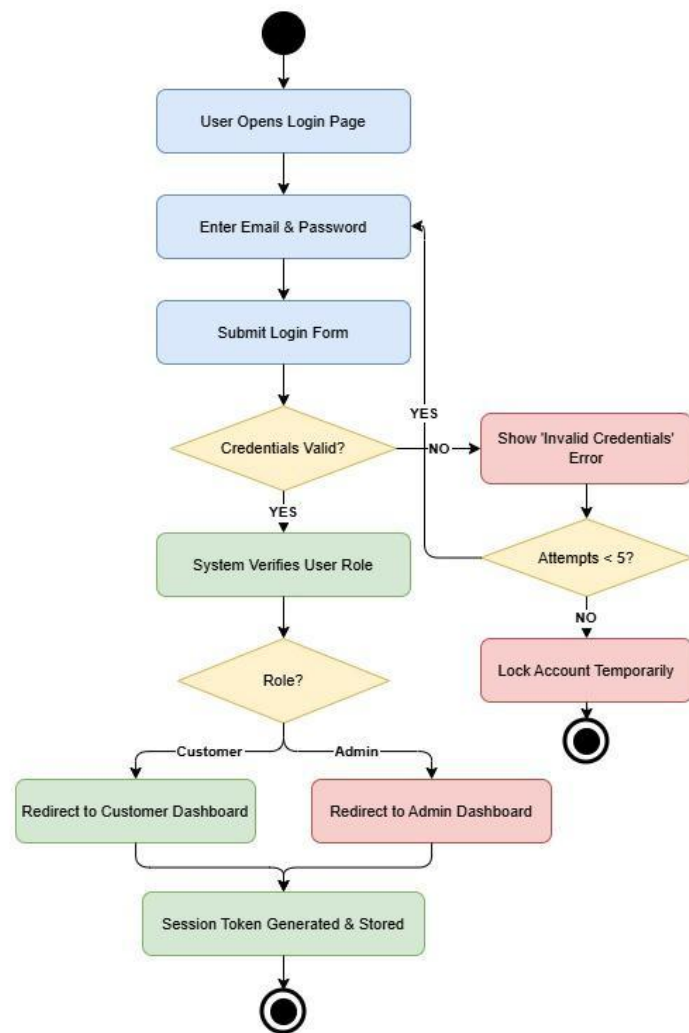
7.5 User Login / Authentication

This activity diagram describes the authentication process for both customers and administrators accessing the Gaming Castle web system, including credential validation, role-based redirection, and account lockout after repeated failed attempts.

Step	Activity	Actor	Decision / Outcome
1	User Opens Login Page	User	–
2	Enter Email & Password	User	–
3	Submit Login Form	User	–
4	Validate Credentials	System	Decision: Credentials Valid? YES → Step 5 NO → Show error
5	Check Failed Attempts	System	Decision: Attempts < 5? YES → Retry NO → Lock account → END
6	Verify User Role	System	–
7	Role-based Redirect	System	Decision: Role? Customer → Customer Dashboard Admin → Admin Dashboard
8	Generate & Store Session Token	System	Postcondition: User is authenticated and redirected

Table 36: Activity Steps – User Login / Authentication

Activity Diagram — User Login / Authentication



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Figure 6: Activity Diagram – User Login / Authentication

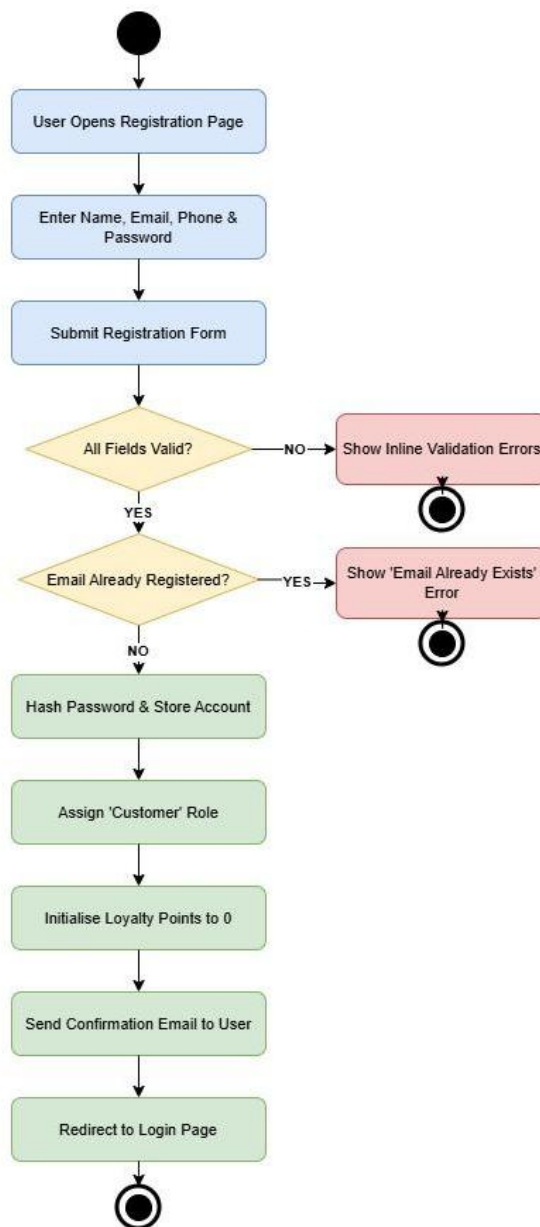
7.6 User Registration

This activity diagram describes the process by which a new user creates an account on the Gaming Castle web system, including form validation, duplicate email checking, account creation, and confirmation email dispatch.

Step	Activity	Actor	Decision / Outcome
1	User Opens Registration Page	User	–
2	Enter Name, Email, Phone & Password	User	–
3	Submit Registration Form	User	–
4	Validate All Fields	System	Decision: All Fields Valid? NO → Show inline errors → END
5	Check Email Uniqueness	System	Decision: Email Already Registered? YES → Show Email Exists error → END
6	Hash Password & Store Account	System	–
7	Assign Customer Role	System	–
8	Initialise Loyalty Points to 0	System	–
9	Send Confirmation Email	System	–
10	Redirect to Login Page	System	Postcondition: New customer account is active

Table 37: Activity Steps – User Registration

Activity Diagram — User Registration



6

Figure 7: Activity Diagram – User Registration

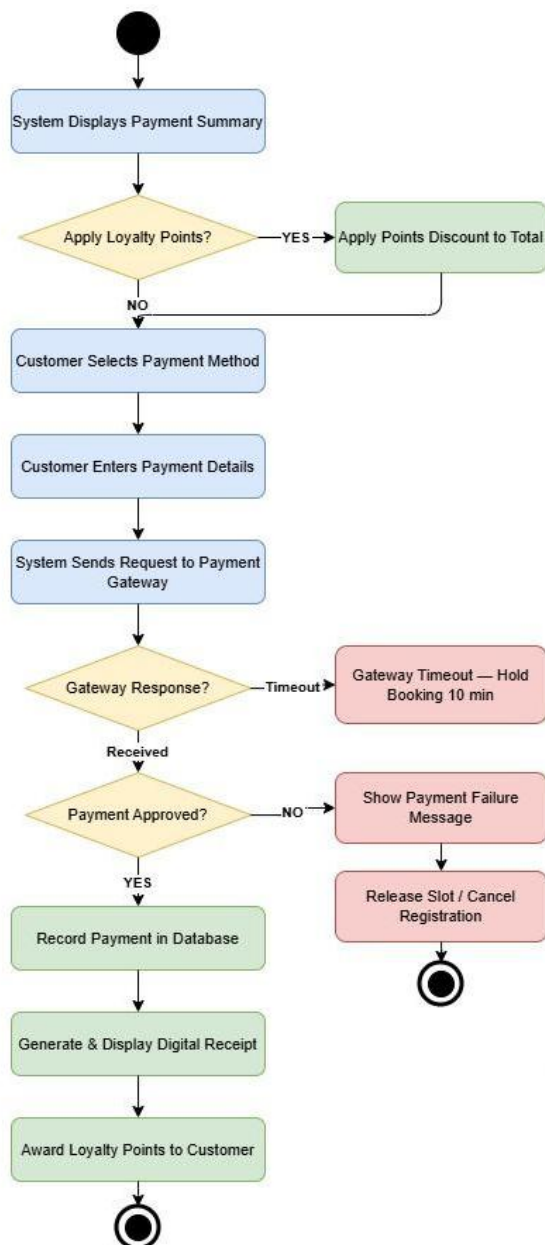
7.7 Payment Processing

This activity diagram describes the end-to-end payment processing workflow within the Gaming Castle system, covering optional loyalty point discount application, payment gateway interaction, receipt generation, and loyalty point award upon successful payment.

Step	Activity	Actor	Decision / Outcome
1	System Displays Payment Summary	System	–
2	Check Loyalty Points Option	System	Decision: Apply Loyalty Points? YES → Apply discount NO → Proceed
3	Customer Selects Payment Method	Customer	–
4	Customer Enters Payment Details	Customer	–
5	Send Request to Payment Gateway	System	–
6	Await Gateway Response	System	Decision: Timeout? YES → Hold booking 10 min Received → Step 7
7	Check Payment Approval	System	Decision: Approved? YES → Step 8 NO → Show failure, release slot → END
8	Record Payment in Database	System	–
9	Generate & Display Digital Receipt	System	–
10	Award Loyalty Points to Customer	System	Postcondition: Payment confirmed, loyalty points updated

Table 38: Activity Steps – Payment Processing

Activity Diagram — Payment Processing



7

Figure 8: Activity Diagram – Payment Processing

7.8 Admin Dashboard Overview

This activity diagram describes the workflow of an administrator interacting with the Gaming Castle admin dashboard, including navigation across key management modules: bookings, tournaments, users, and reports.

Step	Activity	Actor	Decision / Outcome
1	Admin Logs In Successfully	Admin	Precondition: Admin authenticated
2	System Loads Admin Dashboard	System	Displays: Real-time bookings, revenue summary, active sessions
3	Admin Selects Action	Admin	Decision: Manage Bookings Manage Tournaments Manage Users Generate Reports
4a	Manage Bookings	Admin	View / cancel / create bookings
4b	Manage Tournaments	Admin	Create / update / cancel tournaments
4c	Manage Users	Admin	View / deactivate / edit accounts
4d	Generate Reports	Admin	Select report type and date range
5	Action Completed	System	All branches merge here
6	Continue on Dashboard?	Admin	Decision: YES → Refresh dashboard → Step 3 NO → Logout → END

Table 39: Activity Steps – Admin Dashboard Overview

Activity Diagram — Admin Dashboard Overview

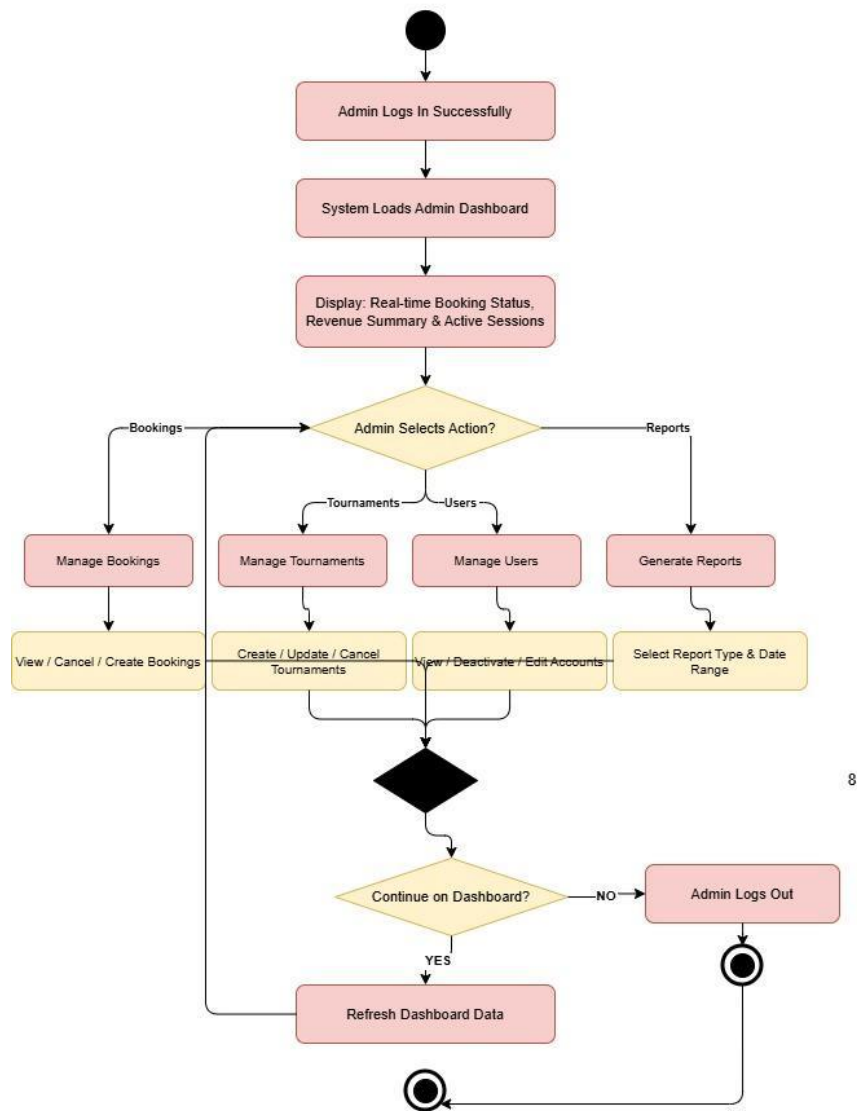


Figure 9: Activity Diagram – Admin Dashboard Overview

Chapter 8 External Interface Requirements

8.1 User Interface Requirements

A clean, intuitive, and responsive web-based user interface shall be provided by the system. The following requirements apply:

- The interface shall be accessible on desktop, tablet, and mobile devices through responsive design (minimum viewport width of 320px).
- The customer-facing interface shall include pages for: Home, Slot Booking, Tournaments, Loyalty Rewards, Profile, and Payment History.
- The admin interface shall include a dashboard with real-time booking status, revenue summary, tournament management, user management, and reporting tools.
- Error messages shall be displayed inline, written in plain English, and accompanied by suggested corrective actions.
- A consistent colour scheme, typography, and iconography aligned with the Gaming Castle brand shall be used by the interface.

8.2 Hardware Interface Requirements

The Gaming Castle system is a web-based application and does not require direct integration with specialised hardware. However, the following hardware dependencies apply:

- A server or cloud hosting infrastructure with a minimum of 2 vCPU, 4 GB RAM, and 50 GB storage is required by the system.
- A modern web browser and stable internet connectivity are required at administrator terminals.
- No direct hardware-level integration with PS5 consoles is required; console allocation is managed logically within the system.

8.3 Software Interface Requirements

- Payment Gateway: A payment service provider shall be integrated via REST API for processing online payments.
- Email Service: An email service shall be integrated for sending booking confirmations, tournament notifications, and password reset emails.
- SMS Notifications: An SMS gateway may be integrated for booking confirmation messages.

- Database: A relational database management system shall be used for persistent data storage.
- Web Server: The system shall be served via a web application server compatible with SpringBoot.

8.4 Communications Interface Requirements

- All client-server communications shall use HTTPS over TLS 1.2 or higher.
- RESTful APIs shall be used for communication between the frontend and backend, with data exchanged in JSON format.
- The system shall operate over standard TCP/IP networks and shall not require any proprietary network protocols.
- WebSocket connections may be used for real-time slot availability updates.

8.5 System Architecture

The Gaming Castle system shall be designed using a microservices-based architecture to improve scalability, maintainability, and flexibility.

Each major module of the system, including User Management, Booking Management, Payment Processing, Tournament Management, and Loyalty Program, shall be implemented as independent microservices.

These microservices shall communicate with each other using RESTful APIs.

Docker containerisation shall be used to package each microservice, ensuring consistent deployment across development, testing, and production environments.

An API Gateway shall be used to handle incoming client requests, manage routing to appropriate services, and enforce security mechanisms such as authentication and authorization.

This architecture provides the following benefits:

- Independent deployment of services
- Improved system scalability
- Easier maintenance and debugging
- Better fault isolation

Chapter 9 Data Requirements

The following key data entities are required for the Gaming Castle system:

Entity	Key Attributes	Description
User	user_id, name, email, phone, password_hash, role, loyalty_points, created_at, updated_at	Stores registered customer and admin account information.
Console	console_id, name, type (PS5), console_status, games_available	Represents a physical PS5 console available for booking.
Booking	booking_id, user_id, console_id, date, start_time, end_time, total_charge, payment_status, booking_type, booking_status	Records all slot booking transactions, both online and walk-in.
Payment	payment_id, booking_id, amount, method (online/cash), payment_status, timestamp	Stores payment records linked to bookings.
Tournament	tournament_id, name, game, date, entry_fee, max_participants, status, created_by	Manages tournament events created by the administrator.
Tournament Registration	registration_id, tournament_id, user_id, payment_id, registered_at	Links customers to tournaments they have registered for.
Loyalty Transaction	transaction_id, user_id, points_awarded, points_redeemed, balance, booking_id, timestamp	Tracks loyalty point earn and redemption history.

Table 40: Data Requirements – Key Entities

All personal data (names, emails, phone numbers) shall be stored in compliance with applicable data protection regulations. Customer passwords shall be stored exclusively as bcrypt hashes.

Chapter 10 Alternative Solutions Considered

Alternative	Description	Reason for Rejection
WhatsApp / Phone-based Booking	Continuation with manual booking via phone calls or WhatsApp messages.	Does not scale; prone to human error; no automated payment or loyalty tracking; rejected in favour of the proposed web system.
Third-party Booking Platform	Adoption of an existing venue booking platform (e.g., Booksy, SimplyBook.me).	Generic platforms do not support gaming-specific features (tournament management, per-console booking, loyalty points); high recurring subscription costs.
Mobile Application (Native)	Development of a native iOS/Android mobile application instead of a web application.	Higher development cost and time; separate codebases are required for iOS and Android; a responsive web application achieves the same goals with a single codebase.
Spreadsheet-based System	Use of Google Sheets or Microsoft Excel for booking and payment tracking.	Not suitable for real-time multi-user access; lacks payment integration and automated notifications; not scalable.

Table 41: Alternative Solutions Considered

Chapter 11 Feasibility Study

11.1 Technical Feasibility

The proposed system is technically feasible. The required skills for building a web application using modern frameworks such as Angular.js (frontend) and Spring Boot (backend) are available within the development team. Relational databases (PostgreSQL), RESTful API design, and cloud hosting platforms are well-established technologies with extensive documentation and community support. Well-documented integration guides are provided by payment gateway APIs (Stripe). No novel or unproven technologies are required.

The use of Docker and microservices architecture further strengthens the technical feasibility of the system. Docker ensures a consistent runtime environment across different stages of development, while microservices enable modular development and scalability. The development team has sufficient knowledge to implement and manage containerised services within the project scope.

An automated tournament management module can be implemented using rule-based logic or simple machine learning techniques. Given the scale of the system, a lightweight approach is sufficient to generate tournament brackets and manage match outcomes without requiring complex infrastructure.

11.2 Economic Feasibility

The system can be developed using open-source frameworks and tools at minimal licensing cost. Cost-effective deployment options are offered by cloud hosting providers (AWS Free Tier). The expected development cost is limited to the time investment of the four-member team within the academic project context. For Gaming Castle, a return on investment is projected to be generated by reducing manual overhead, increasing booking efficiency, and attracting a broader customer base through online accessibility and tournament events.

11.3 Operational Feasibility

The system is operationally feasible. The administrator interface is designed to be intuitive and requires minimal training. The customer-facing interface follows familiar web conventions for booking and payment. Internet connectivity is available at the Gaming

Castle premises, and web-based services are routinely used by the target users. Customer engagement and retention are expected to be increased through the loyalty programme and tournament features.

11.4 Schedule Feasibility

The project is planned to be completed within the academic semester. Four members with defined responsibilities constitute the team. The development is structured into sprints covering requirements analysis, system design, frontend development, backend development, integration testing, and final deployment. Based on the project scope and team capacity, the schedule is considered achievable within the allocated timeframe.

Chapter 12 Risk Analysis

The following Risk Register documents identified project risks, their likelihood, impact, overall risk level, and proposed mitigation strategies.

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
R-01	Payment gateway integration failure or delays	Technical	Medium	High	High	Multiple payment gateway options shall be researched early; a mock payment module shall be developed for testing.
R-02	Scope creep due to evolving client requirements	Requirements	High	Medium	High	A formal change request process shall be established; client sign-off on requirements shall be obtained before development.

R-03	Team member unavailability due to illness or academic workload	Resource	Medium	Medium	Medium	All work shall be thoroughly documented; knowledge shall be distributed across the team; task backups shall be maintained.
R-04	Security vulnerabilities in the web application	Technical	Medium	High	High	An OWASP Top 10 assessment shall be conducted; HTTPS and input validation shall be enforced; peer code reviews shall be performed.
R-05	Project timeline overrun	Schedule	Medium	High	High	Agile sprint planning shall be used; weekly progress reviews shall be conducted; core features shall be prioritised for MVP delivery.
R-06	Low user adoption post-deployment	External	Low	Medium	Low	The client and target users shall be involved in UAT; a user guide and onboarding support shall be provided.

R-07	Third-party API deprecation or changes	Technical	Low	High	Medium	Stable, well-maintained APIs shall be selected; abstraction layers shall be implemented to facilitate API replacement.
R-08	Data loss due to server failure	Technical	Low	High	Medium	Automated daily database backups with off-site storage shall be implemented; recovery procedures shall be tested.

Table 42: Risk Register

Chapter 13 References

- [1] IEEE, "IEEE Recommended Practice for Software Requirements Specifications," IEEE Std 830-1998, 1998.

- [2] ISO/IEC, "Systems and software engineering – Systems and software Quality Requirements and Evaluation (SQuaRE) – System and software quality models," ISO/IEC 25010:2011, 2011.

- [3] OWASP Foundation, "OWASP Top Ten," 2021. [Online]. Available: <https://owasp.org/www-project-top-ten/>. (accessed Mar. 11, 2026).

- [4] W3C, "Web Content Accessibility Guidelines (WCAG) 2.1," 2018. [Online]. Available: <https://www.w3.org/TR/WCAG21/>. (accessed Mar. 11, 2026).

Client Requirement Sign-Off

The client and project supervisor must review and approve the scope, requirements, and features documented in this SRS before the project proceeds to the design phase. Any changes to the approved requirements after this sign-off must be submitted as a formal Change Request and re-approved by both parties.

Client Representative Sign-Off

Name: Lathurshigan

Title: Owner

Organisation: Gaming Castle

Date: 16/04/2026

Signature: 

Project Supervisor Sign-Off

Name: _____

Date: _____

Signature: _____